



Comprehensive Genomic Analysis

A unified platform for **germline variant interpretation**, genome visualisation, and **signed clinical reporting** — from an annotated VCF to a version-pinned, traceable report.

C Rare-disease & inherited

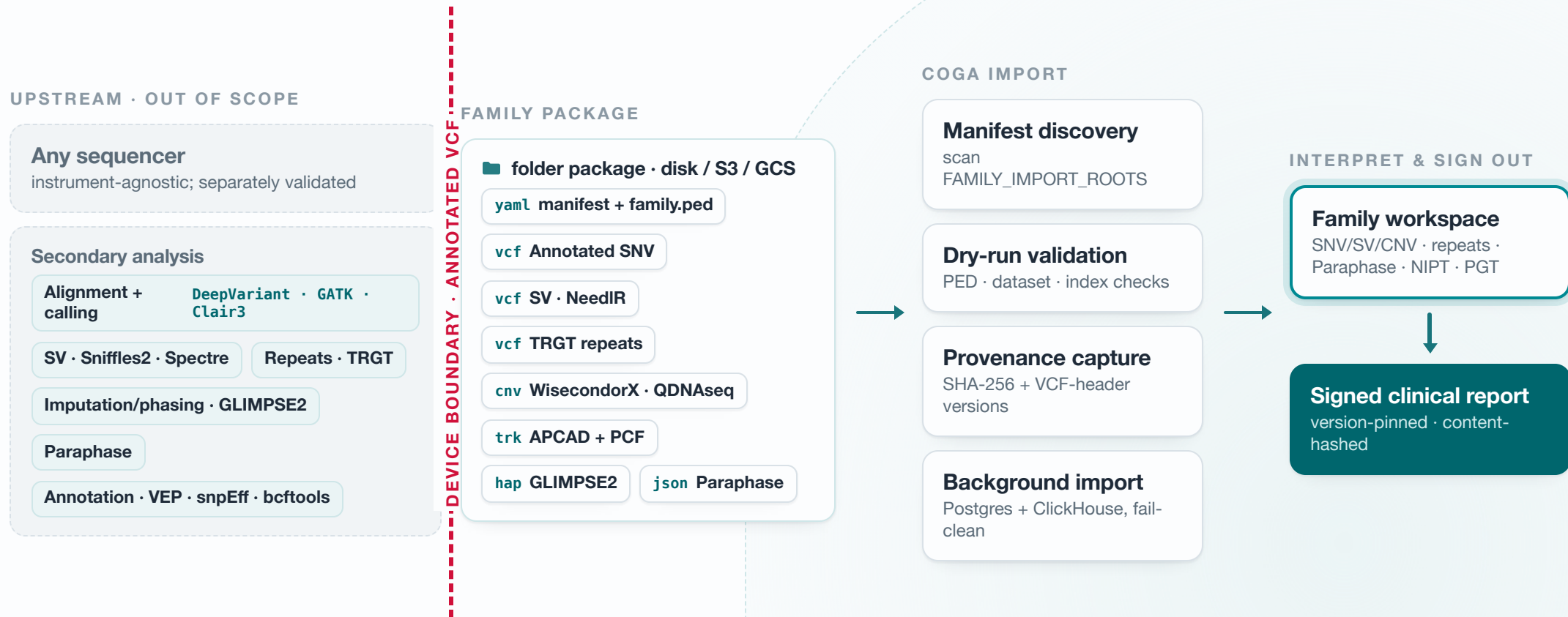
G Reproductive genomics

A Interpretation, QC & sign-out

In-house IVD · **IVDR Art 5(5)** · CMGG · ISO 15189 · FastAPI · React · Postgres · ClickHouse

Instrument-agnostic ingestion

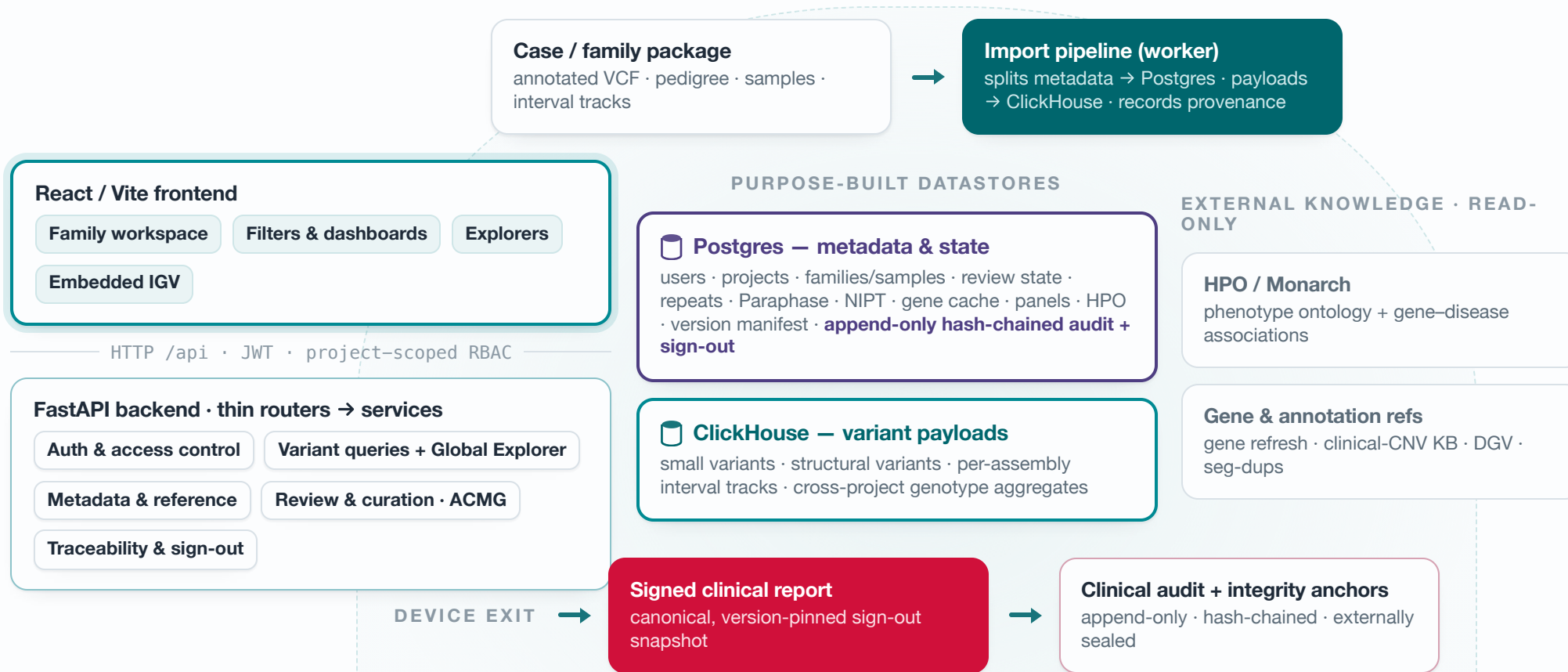
Whatever the upstream sequencer, CoGA ingests the same annotated-VCF-plus-tracks family package — so the regulated device begins only at the validated digital input.



- **Device boundary:** everything left of the annotated VCF (assay → sequencing → calling → annotation) is separately validated and out of scope. Upstream tool/DB versions are read from the VCF headers and frozen per family. CoGA names no instrument — it consumes the same package regardless of platform.

One case, end to end

An imported case package on one side, two purpose-built databases in the middle, external knowledge feeding in — and a tamper-evident signed report on the other.



- **In-house IVD · IVDR Art 5(5)** — Postgres is authoritative for metadata & state, ClickHouse for variant payloads. External services are read-only reference refreshes; all data in the platform is synthetic in non-clinical use.

From reviewed variants to a signed, traceable report

Every reported result is locked to the exact software, annotation and reference versions that produced it — then sealed in a tamper-evident audit trail.

1 · REVIEW

ACMG/AMP classification
semi-automatic 5-class score,
recomputed server-side; every criterion
overridable

Tags & notes
the report tag selects variants;
who/when + analyst note

Evidence snapshot
frozen at each save: annotation-set-
hash · ClinVar · timestamp

2 · GATING

Classification-drift
frozen vs current annotation-set-
hash — 409 unless acknowledged

Sample-QC acknowledgement
sex · relatedness · Mendelian ·
NIPT + swap checks — blocks
unless acknowledged with a
reason

3 · SIGN-OUT

Frozen versioned snapshot
SHA-256 content-hashed, append-only;
amendments create a new version —
prior versions never mutate

Bound to three version axes

sw build + git_sha

ann manifest + set-hash

ref assembly

4 · OUTPUTS

Signed report
variant sections · gene & HPO context ·
provenance footer (VEP · ClinVar ·
gnomAD · HPO ...) · PDF export

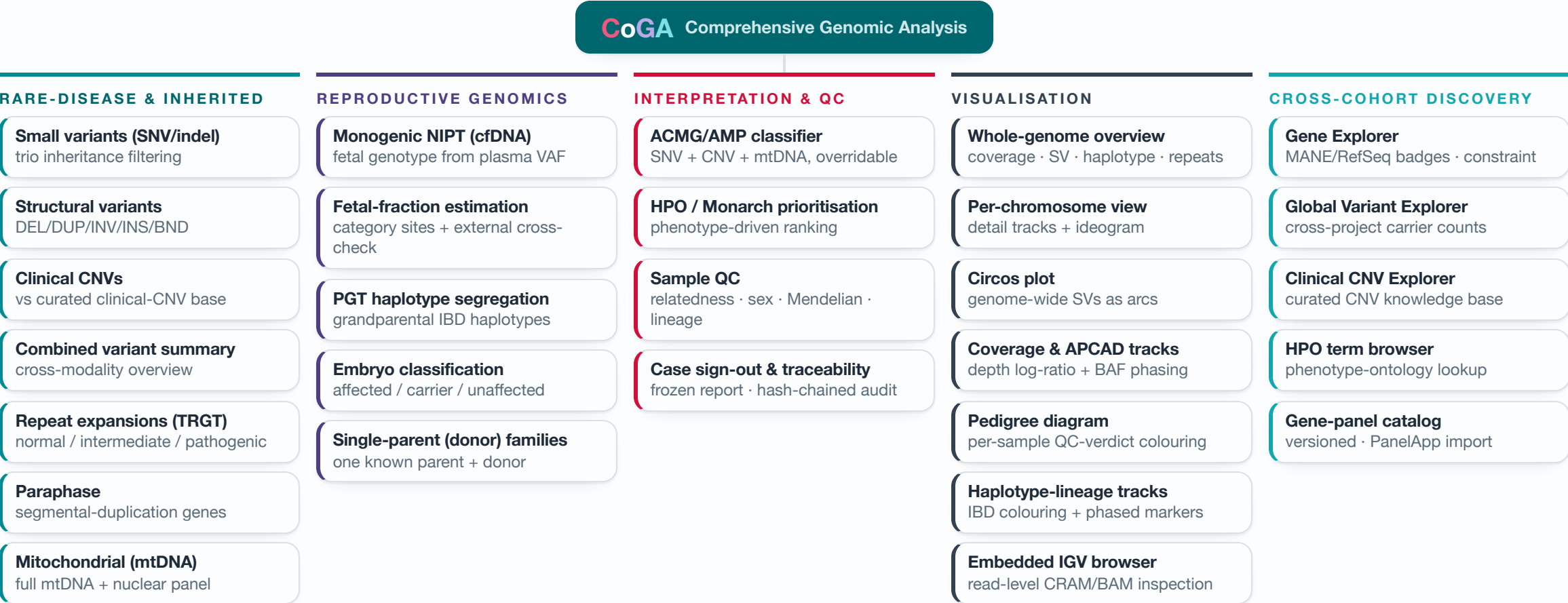
Hash-chained audit
append-only before/after log; each
entry's row_hash links to the prior —
edits/reordering become detectable

seal Ed25519 integrity anchors

In-house IVD · IVDR Art 5(5). The chain is tamper-evident against a database-only adversary between retained anchors, not tamper-proof. ACMG/AMP output is decision support — confirmed by a qualified clinical scientist before clinical use.

What CoGA does

A germline variant-interpretation platform for rare-disease and reproductive genomics — no somatic-oncology, HLA or transplant menus.



● Per-data-type family views surface only when that data is present. Scope is germline rare-disease + reproductive genomics — operated as an in-house IVD under IVDR Art 5(5).

Integrated knowledge for clinical interpretation

The ontologies, population and clinical-variant databases, transcript references and genomic-context tracks CoGA brings together behind every call.

PHENOTYPE & DISEASE

- **HPO** · phenotype ontology + closure
- **Monarch Initiative** · gene–disease–phenotype graph
- **OMIM** · Mendelian gene–disease
- **Orphanet** · rare-disease associations
- **ClinGen** · validity · dosage · ACMG calibration
- **GenCC** · multi-submitter validity
- **PanelApp** · curated disease gene panels

POPULATION & CONSTRAINT

- **gnomAD** · allele frequency + constraint (pLI · LOEUF · missense-Z)
- **dbNSFP** · bulk gene annotation & constraint

CLINICAL VARIANT KNOWLEDGE

- **ClinVar** · curated clinical assertions
- **In-silico predictors** · CADD · REVEL · SpliceAI · AlphaMissense
- **ACMG/AMP guidelines** · Richards 2015 · Tavtigian/ClinGen points · Riggs CNV · mtDNA

TRANSCRIPTS & REFERENCE

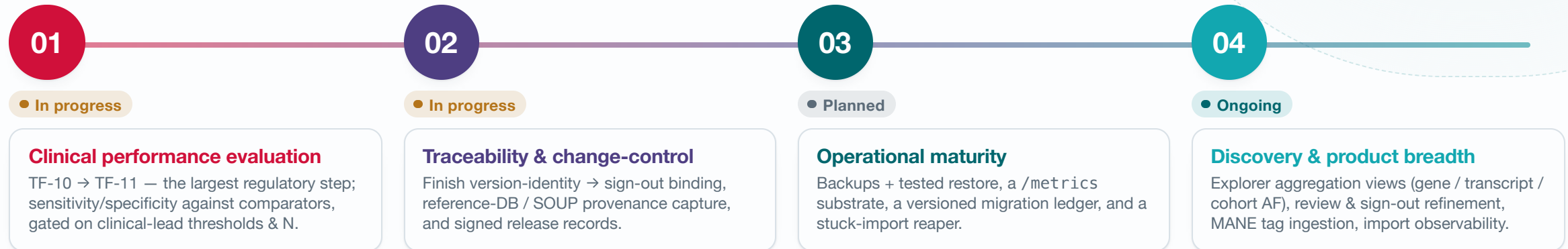
- **MANE** Select / Plus Clinical · reference transcripts
- **RefSeq / NCBI Gene** · accessions + summaries
- **Ensembl** · ids · biotype · canonical
- **HGNC** · official gene nomenclature
- **GENCODE** · transcript annotation set
- **Reference genomes** · GRCh38 · T2T-CHM13 · GRCh37
- **UCSC** · gene tables + cytobands

GENOMIC CONTEXT TRACKS

- **DGV** · common structural/CNV context
- **Segmental duplications / LCRs** · FP-prone loci
- **Blacklist / exclusion regions** · artifact zones
- **Clinical CNV knowledgebase** · known clinical CNVs

Where CoGA is heading

Direction is weighed for its regulatory consequence, not engineering merit alone — validation and traceability lead, product breadth follows.



- **Regulatory framing:** in-house IVD under IVDR Art 5(5) (CMGG, ISO 15189). Engineering work is weighed for GSPR / ISO 14971 / IEC 62304 / ISO 27001 / GDPR consequence. See the technical file & improvement workplan.